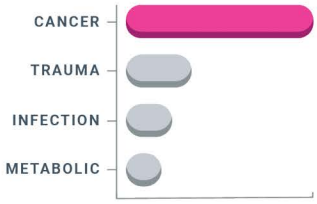
OncoK9 Resources

Early Cancer Detection in Dogs Starts with a Simple Blood Draw



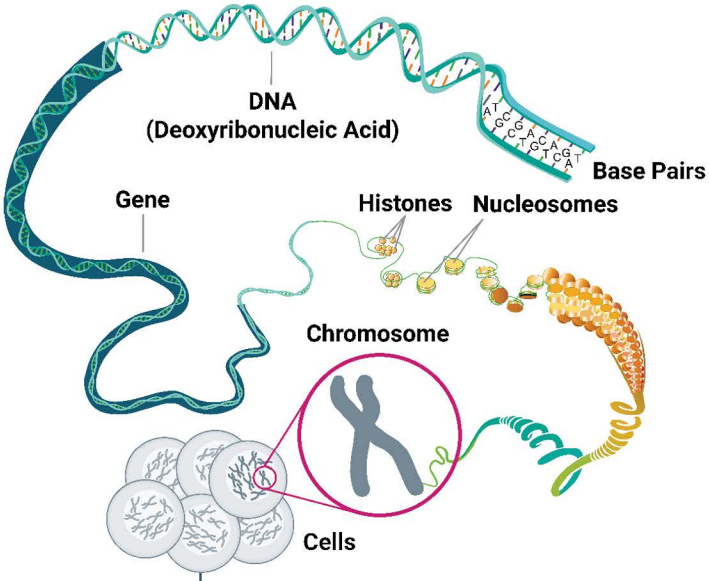
Cancer Is by Far the
#1 Cause of Death in Dogs

6 Million New Cancer Cases Per Year in Dogs

Older dogs and certain breeds have a higher lifetime risk of cancer and/or have a higher risk of being diagnosed at an earlier age. Cancer in dogs is often diagnosed at an advanced stage, when treatment options are limited, and chances of long-term control or cure are low.

> People get mammograms, colonoscopies, and other cancer screening tests to help detect cancer early. Now, our dogs have OncoK9, a cancer screening test developed specifically for them, ushering in a new era of preventive care for pets.

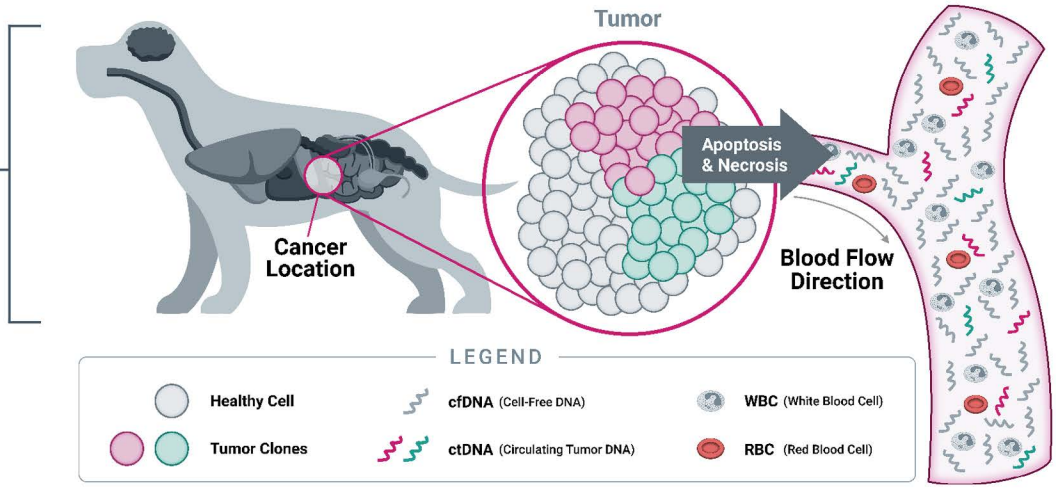
Lifetime Risk of Cancer
1 in 3 Dogs



Cancer Is a Disease of the Genome[1]

Each cell in a dog's body contains billions of DNA "letters" that make up the dog's genome. Cancer develops as a result of the successive accumulation of genomic alterations (DNA mutations) over time. All cells release DNA fragments into the bloodstream. DNA released from cancer cells contains specific genomic alterations associated with cancer that can be detected by liquid biopsy with the **OncoK9** test.

OncoK9 indicates whether cancer is currently present in the body; it does not indicate risk for future cancer development.



OncoK9: The Liquid Biopsy Test for Dogs

As a first-in-class multi-cancer early detection (MCED) test, OncoK9 employs cutting-edge genomic analysis that leverages next-generation sequencing (NGS) technology and proprietary bioinformatics algorithms, empowering veterinarians to provide superior care to canine patients.

Simple Blood Draw

- Can be drawn in clinic same day
- No fasting required
- No sample processing required in clinic
- Validated in cancer-diagnosed dogs and in cancer-free dogs with a range of other medical conditions

Multi-Cancer Coverage

- Proven ability to detect 30 distinct cancer types, with high detection rates in many of the most common cancers in dogs
- Cancer signal origin prediction in a subset of hematological malignancies

Cutting-Edge Technology

- Uses next-generation sequencing (NGS) technology
- No refrigeration or freezing of the sample; shipped overnight at ambient temperature
- Low false positive rate

Early Detection

- Identifies cancer signal in DNA extracted from a blood sample
- Earlier cancer detection before the onset of clinical signs may be possible

OncoK9 Multi-Cancer Detection

List of cancer types detected in the CANDiD (CANcer Detection in Dogs) study by histologic type and/or anatomic location

3 of the most aggressive canine cancers shown in **pink** | *8 of the most common canine cancers shown in* **bold**

A
Abdominal Cavity
Anal Sac Adenocarcinoma

B
Bile Duct
Bone, Osteosarcoma
Brain

C
Chondrosarcoma

E
Ear Canal

H
Heart Base
Hemangiosarcoma
Histiocytic Sarcoma

K
Kidney

L
Leukemia, Acute Lymphoid (ALL)
Leukemia, Chronic Lymphoid (CLL)
Liver
Lung
Lymphoma, Indolent
Lymphoma, Intermediate to Large Cell

M
Malignant Melanoma
Mammary Gland Carcinoma
Mast Cell Tumor

N
Nasal Cavity and Paranasal Sinuses

O
Oral Cavity

P
Peripheral Nerve Sheath

S
Salivary Gland
Skin
Soft Tissue Sarcoma
Stomach

T
Thyroid
Transmissible Venereal Tumor

U
Urinary Bladder / Urethra

OncoK9 Test Performance

Cancer Type (Canine)	Screening Method	Detection Rate	False Positive Rate*
3 of the most aggressive canine cancers	OncoK9	85%	1.5%
8 of the most common canine cancers	OncoK9	62%	1.5%
All cancers in CANDiD Study	OncoK9	55%	1.5%

**False Positive Rate of 1.5% corresponds to a Specificity of 98.5%*

Performance of Commonly Used Human Cancer Screening Tests[2-15]

OncoK9 provides cancer screening performance that meets and, in many cases, exceeds established human standards

Cancer Type (Human)	Screening Method	Detection Rate	False Positive Rate^
Breast	Digital mammogram	77 - 95%	3 - 6%
Cervical	Pap test	68 - 94%	25 - 74%
Prostate	PSA level^^	21 - 86%	9 - 67%
Lung	Low dose CT	59 - 100%	1 - 74%
Colon	Stool DNA-based test^^^	92 - 93%	13 - 15%

^ False Positive Rate = 1 – Specificity *^^ PSA level (>4ng/mL)* *^^^ Cologuard*

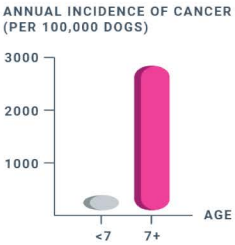
OncoK9 Test Interpretation: Probability of Cancer or Cancer-Free

Clinical use case	Intended use population	Probability that a patient with an OncoK9 *Cancer Signal Detected* result has cancer (PPV)	Probability that a patient with an OncoK9 *Cancer Signal Not Detected* result does not have cancer (NPV)
Screening	Higher risk of cancer due to age and/or breed	76 - 80%	95 - 96%
Aid-in-diagnosis	Cancer suspected based on clinical presentation	94 - 97%	68 - 84%

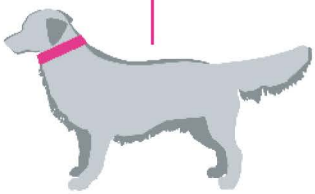
Estimated ranges for positive predictive value (PPV) and negative predictive value (NPV) calculated using a test sensitivity of 54.7% and specificity of 98.5% (based on 8-10% prior probability in the screening use case, and 30-50% prior probability in the aid-in-diagnosis use case).

OncoK9 Clinical Use Cases

OncoK9 is recommended as an
annual screening test
for dogs at higher risk of cancer[16]



9x Higher Risk of Cancer **Over Age 7**



Higher Risk of Cancer in **Certain Breeds**

- Beagle
- Bernese Mountain Dog
- Boxer
- Flat-Coated Retriever
- French Bulldog
- German Shepherd
- Golden Retriever
- Labrador Retriever
- Rhodesian Ridgeback
- Rottweiler
- Siberian Husky
- Giant Breeds *(including Great Dane, Irish Wolfhound, Mastiff, and Scottish Deerhound)*

OncoK9 is recommended as an
aid-in-diagnosis test
for dogs in which cancer is suspected

Cancer Site Is Suspected but **Challenging to Access**

Cancer Is Suspected but Site Is **Not Clinically Evident**

OncoK9 Test Workflow

i	Blood collection at clinic	v	PetDx results review (lab director & PetDx veterinary team)
ii	Shipment to lab	vi	Veterinarian receives OncoK9 report
iii	DNA extraction & sequencing	vii	A board-certified Small Animal Internal Medicine specialist from PetDx provides consultation on result interpretation and diagnostic workup for positive results
iv	Bioinformatics analysis	viii	Veterinarian communicates OncoK9 test results & next steps to pet owner

Easy Integration Into Your Practice

How do I order OncoK9 tests?

Once you have a PetDx account, you may order OncoK9 tests and receive results through our Veterinary Portal at **vet.petdx.com**. It is intuitively designed with veterinarians and veterinary staff in mind.

How do I receive OncoK9 support?

Check out our Veterinary Support site at **support.petdx.com** for a list of FAQs and resources or to submit customer support requests. Specialist consultations are available to discuss cancer treatment options. Contact our Customer Support Team at **(833) 464-7297** or **help@petdx.com**.

To learn about the rapidly growing number of patients that have benefitted from OncoK9, please visit **petdx.science/case-studies**.

For a list of references 1-16 used in this document, please visit ***OncoK9.info/Veterinary-Overview***